
UNIT 4 NATURAL RESOURCES

Structure

- 4.0 Objectives
- 4.1 Introduction
 - 4.1.1 Knowledge of Natural Resources
- 4.2 Land and Soils
 - 4.2.1 Land Utilisation in India
 - 4.2.2 Issue of Land acquisition
 - 4.2.3 Need for a Comprehensive Land-Use Policy
 - 4.2.4 Soils
- 4.3 Cropping Pattern in India
 - 4.3.1 Crop Diversity
 - 4.3.2 Future Cropping Pattern in India
- 4.4 Water Resources
 - 4.4.1 Water Issues and Solutions
 - 4.4.2 National Law on Water
- 4.5 Biodiversity
 - 4.5.1 The Biological Diversity Act, 2002
- 4.6 Forest Resources
 - 4.6.1 Present Position
 - 4.6.2 National Forest Policy
- 4.7 Mineral Resources
 - 4.7.1 Features of Minerals
 - 4.7.2 New Mineral Policy, 2008
 - 4.7.3 Acquiring Mineral Sources Abroad
- 4.8 Allocation of Natural Resources
- 4.9 Environment and Economic Development
 - 4.9.1 Environmental Protection in India
 - 4.9.2 National Environment Policy, 2006 (NEP)
- 4.10 Let Us Sum Up
- 4.11 Term-end Exercises
- 4.12 Key Words
- 4.13 References
- 4.14 Answers or Hints to Check your Progress Exercises

4.0 OBJECTIVES

As you go through this Unit, you will begin to appreciate the:

- importance of natural resources in the growth process of an emerging economy like India;
- significance of policies to create an eco-system for optimum utilisation of natural resources;

- the present status of knowledge and availability of different natural resources in India;
- policy framework guiding the use of natural resources in India; and
- need for a comprehensive resources-use policy and suggested outline for such a policy.

4.1 INTRODUCTION

The availability of resources, their utilisation, and the capacity to mobilise additional resources, determine an economy's ability to overcome the constraints on its development. As the world's resources grow scarce, India faces an uphill task to secure the resources required for its progress and future. On the demand side, growth in population and urbanisation is increasing competition for all kinds of resources. Meanwhile, resource supplies are being constrained by geopolitical factors and inadequate complementary infrastructure to support resource exploration and generation at a required pace. As a result of this double-edged scarcity, concerns are growing over the availability and prices of resources. Yet, approached imaginatively, the quest for efficiency of resource use can become an important source of economic growth and job creation.

As in most other Emerging Economies (EEs), a large proportion of population of India is dependent upon agriculture and other primary activities which consist in direct exploitation of natural resources. The natural resources of India, as the following survey indicates, are varied and provide an adequate basis for building a diversified modern economy.

4.1.1 Knowledge of Natural Resources

Information on stock of natural resources is much better in case of India than in most EEs. Basic resource survey agencies have been in existence for a century or more and systematic surveys and investigations of resources have been undertaken by them.

Work on mapping and surveys of natural resources has expanded greatly under the Five-Year Plans. Some of the institutional arrangements put in place in this regard are:

- The older agencies, such as the Survey of India and the Geological Survey, have been expanded to enable them to undertake larger programmes of work, and new agencies have been created to undertake specialised tasks or surveys in fields which had not been covered earlier.
- The *Indian Bureau of Mines* was established in 1950 to undertake an economic assessment of natural resources and to formulate programmes of development.
- The *Oil and Natural Gas Commission*, established in the late 1950's to undertake exploration and development of petroleum resources, has been responsible for major discoveries of oil and natural gas in Assam, Gujarat and Bombay High.
- The *Central Water and Power Commission* (now bifurcated into two), established in 1945, has the responsibility for the coordination of hydrological investigations and assessment of natural resources.

- A *Soil Land Use Survey* has been established under the Indian Council of Agricultural Research; systematic soil surveys are being undertaken by it in cooperation with the State Departments of Agriculture.
- The newly introduced Forest Survey, the State of Forest Report which is published at regular intervals, etc. are regular source of information and data for policy-makers.
- The national laboratories and institutes of scientific research also undertake studies relating to evaluation for utilisation of natural resources.

The experience under the Plans has demonstrated the need for further expansion or acceleration of work in some directions: a greater use of modern survey and mapping techniques and an adequate economic assessment of natural resources. Investigations of soil fertility and assessment of groundwater reserves will need to be improved to provide data for an intensive use of chemical fertilizers and efficient water management, which are essential for the successful use of the new high-yielding varieties of seeds or other programmes of intensive cultivation. Modern survey techniques, such as aerial photography, aero-magnetic survey and remote-sensing, which have advanced very rapidly in recent years, but which are not used adequately by the Indian survey agencies will need to be used much more. The use of these techniques reduces the time requirements of surveys and preparation of maps; makes it possible to survey inaccessible areas and helps in the location or estimation of reserves of minerals and other natural resources.

4.2 LAND AND SOILS

India measures 3,214 kms. from north to south and 2,933 kms. from east to west with a total land area of 32,87,782 sq. kms and a coastline measuring 7516.5 kms., plus 1,197 islands. It is the seventh largest *landowner* in the world after Russia, Canada, China, the U.S.A., Brazil and Australia in that order. In brief, India is a vast country and has a considerable strategic significance on account of its location, size, and economic resources. Standing at the heart of the Indian Ocean, the country is in a much better position than any other in the area to control the Indian Ocean routes, most of which touch the Indian ports. The air routes between Europe, West Asia, Africa and East Asia, South-East Asia and Japan also pass through India. It gives India an advantage in terms of international mobility of persons and commodities.

4.2.1 Land Utilisation in India

The available land, on the basis of its use, can be classified into two parts, *viz.*, (1) Agricultural land, and (2) Non-agricultural land.

A) Agricultural Land

It includes net sown area, current fallows and land under miscellaneous tree crops and groves. Agricultural land in India totals a little over 50 per cent of the total geographical area in the country. This is the highest among the large or medium-sized countries of the world, indicating -

- the influence of favourable physical factors such as large area, the extent of plains and plateaus and a very small extent of arid areas (about 15.8 per cent of the country's geographical area is arid.)
- the extension of cultivation to a large proportion of the cultivable land.